SECTION A

(25 *marks*)

1. Evaluate

(a) $\lim\limits_{x \to 3} \dfrac{\sqrt{x-1}-\sqrt{2}}{9-x^2}$. **(3 marks)**

(b) $\lim\limits_{x \to -\infty} \dfrac{4x+\sqrt{x^2-3}}{x}$. **(3 marks)**

(c) $\lim\limits_{x \to 4^+} \dfrac{16-x^2}{|x-4|}$. **(3 marks)**

2. Find the derivatives of the following functions

(a) $y = \ln\left(x^2\sqrt{x-1}\right)$. **(4 marks)**

(b) $y = 4\sin\left(\dfrac{e^x}{\ln x}\right)$. **(4 marks)**

3. A function *f* is defined by $f(x) = 3x^2 - x^3$.

(a) Find the coordinates of the stationary points and determine the nature of the points. **(3 marks)**

(b) Given that the point (1,2) lies on the curve, show that it is the point of inflection. Hence, sketch the graph of $f(x)$. **(5 marks)**

SECTION B

(75 *marks*)

1. Given a complex number $z=\dfrac{(1-i)^2}{3+i}$.

(a) Express z in Cartesian form $a+bi$, where a and b are real numbers.
(2 marks)

(b) Hence, find the modulus and argument of z. **(3 marks)**

2. Determine the solution set of the following inequalities

(a) $\dfrac{1}{3-x}\geq\dfrac{2}{x-7}$. **(6 marks)**

(b) $\left|\dfrac{x}{x-4}\right|<1$. **(7 marks)**

3. Given that $f(x)=e^{3x}+2$, $x\in\mathbb{R}$.

(a) Find $f^{-1}(x)$. **(3 marks)**

(b) Sketch the graphs of f and f^{-1} on the same axes.
Hence, state the domain and range of f and f^{-1}. (**6 marks**)

4. (a) Given $f(x)=\ln(2x-1)$ and $g(x)=2e^{-x}+1$.
Find $g\circ f(x)$, hence evaluate $g\circ f(4)$. **(6 marks)**

(b) Find the domain and range of $f(x)=3\sin 2x$, $0\leq x\leq 2\pi$,
Hence, sketch the graph of $f(x)$. **(3 marks)**

5. (a) (i) If g is a function with $g'(2)=3$, find $\lim_{x\to 2}\frac{g(x)-g(2)}{\sqrt{2x}-2}$. **(3 marks)**

(ii) Evaluate $\lim_{x\to 2}\frac{1+x^2}{(x-2)(x+1)}$. **(3 marks)**

(b) A function *f* is defined as follows

$$f(x)=\begin{cases} x^3-3 & , \quad x\le -1 \\ a+(x+3)^2 & , \quad -1<x\le 3 \\ bx+4 & , \quad 3<x\le 6 \\ 8 & , \quad x>6 \end{cases}$$

(i) If *f* is continuous at $x=-1$ and $x=3$, find the values of *a* and *b*. **(5 marks)**

(ii) Determine the values of *b* such that *f* is not continuous at $x=6$. **(3 marks)**

6. (a) Given the parametric equation

$$x=4t+\frac{3}{t}, \qquad y=4t-\frac{3}{t} \quad \text{where} \quad t\ne 0.$$

(i) Show that $\frac{dy}{dx}=1+\frac{6}{4t^2-3}$. **(5 marks)**

(ii) Find the value of $\frac{d^2y}{dx^2}$ when $t=1$. **(4 marks)**

(b) Given that $f(x)=\frac{1}{\sqrt{5x+36}}$.

Show that $\frac{f''(x)+f'(x)}{\sqrt{5x+36}}=\frac{75}{4}[f(x)]^6-\frac{5}{2}[f(x)]^4$. **(6 marks)**

7. Water is pumped at a constant rate of 180 cm^3s^{-1} into an empty inverted conical container of radius 30 cm and height 75 cm.

(a) Show that the rate of change of the height of water in the container is $\frac{dh}{dt} = \frac{1125}{\pi h^2}$, where h is the height of water in centimeters at time, t, seconds. **(5 marks)**

(b) Calculate the rate of change of the radius of water in the container at the instant the radius of the surface of water is one third of the radius of the container.
Give your answer in terms of π. **(3 marks)**

(c) Find the time taken in minutes to fill the container completely. Give your answer in 3 significant figures. **(2 marks)**

END OF QUESTIONS PAPER

ANSWER

SECTION A

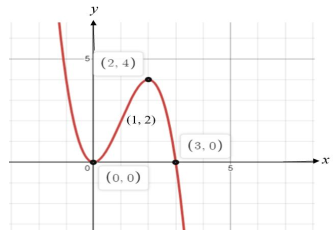
1. (a) $-\dfrac{1}{12\sqrt{2}}$ (b) 3 (c) -8

2. (a) $\dfrac{5x-4}{2x(x-1)}$ (b) $\dfrac{dy}{dx}=4\cos\left(\dfrac{e^x}{\ln x}\right)\bullet\dfrac{xe^x\ln x-e^x}{x(\ln x)^2}$

3. (a) Stationary points are (0, 0) and (2, 4)

Minimum point : (0, 0)
Maximum point : (2, 4)

(b)



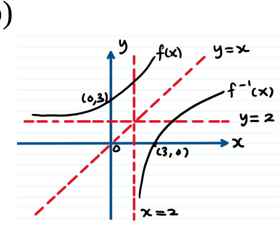
SECTION B

1. (a) $z=-\dfrac{1}{5}-\dfrac{3}{5}i$ (b) $|z|=\dfrac{\sqrt{10}}{5}$ or 0.6325

$Arg(z)=-1.8925$ rad

2. (a) Solution set : $\left\{x : x<3 \text{ or } \dfrac{13}{3}\le x<7\right\}$ (b) Solution set : $\{x: x<2\}$

3. (a) $f^{-1}(x)=\dfrac{1}{3}\ln(x-2)$ (b)

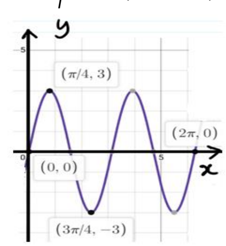


$D_f=(-\infty,\infty)$
$R_f=(2,\infty)$
$D_{f^{-1}}=(2,\infty)$
$R_{f^{-1}}=(-\infty,\infty)$

4. (a) $g(f(x))=\dfrac{2x+1}{2x-1}$, $x\ne\dfrac{1}{2}$

$g(f(4))=\dfrac{9}{7}$

(b) $D_f=[0,2\pi]$, $R_f=[-3,3]$



5. (a) (i) 6

(ii) $\lim\limits_{x\to 2}\dfrac{1+x^2}{(x-2)(x+1)}$ does not exist

(b) (i) $a=-8,\ b=8$

(ii) $b\ne\dfrac{2}{3}$

6. (a)(ii) -48

7. (b) $\dfrac{dr}{dt}=\dfrac{18}{25\pi}$ cm/s (c) $t=6.54$ minutes

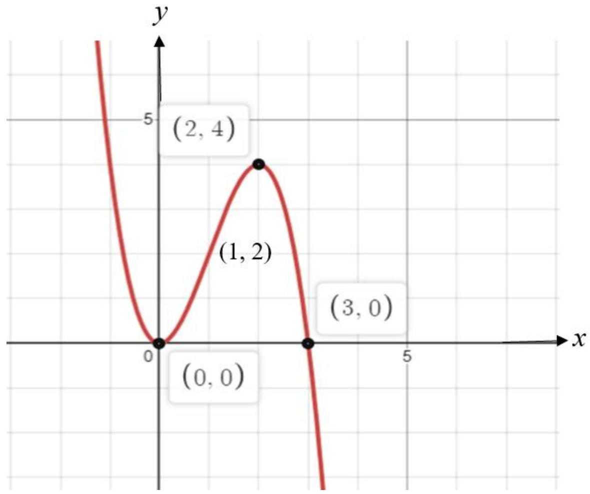
SECTION A

(25 *marks*)

NO	ANSWER SCHEMES	REMARKS
1(a)	$\lim_{x\to 3}\frac{\sqrt{x-1}-\sqrt{2}}{9-x^2}=\lim_{x\to 3}\frac{\sqrt{x-1}-\sqrt{2}}{9-x^2}\times\frac{\sqrt{x-1}+\sqrt{2}}{\sqrt{x-1}+\sqrt{2}}$	**K1** (multiply with its conjugate)
	$=\lim_{x\to 3}\frac{x-3}{(3-x)(3+x)\left(\sqrt{x-1}+\sqrt{2}\right)}$	
	$=\lim_{x\to 3}\frac{-(3-x)}{(3-x)(3+x)\left(\sqrt{x-1}+\sqrt{2}\right)}$	
	$=\lim_{x\to 3}\frac{-1}{(3+x)\left(\sqrt{x-1}+\sqrt{2}\right)}$	
	$=\frac{-1}{(3+3)\left(\sqrt{3-1}+\sqrt{2}\right)}$	**K1** (simplify and substitute *x*=3)
	$=-\frac{1}{12\sqrt{2}}$	**J1**
(b)	$\lim_{x\to -\infty}\frac{4x+\sqrt{x^2-3}}{x}=\lim_{x\to -\infty}\frac{\frac{4x+\sqrt{x^2-3}}{x}}{\frac{x}{x}}$	**K1** (divide by *x*)
	$=\lim_{x\to \infty}\frac{\frac{4x}{x}+\frac{\sqrt{x^2-3}}{-\sqrt{x^2}}}{1}$	**K1** (simplify and substitute value)
	$=\lim_{x\to \infty}4-\sqrt{1-\frac{3}{x^2}}$	
	$=4-\sqrt{1-0}$	**J1**
	$=3$	
1(c)	$\lim_{x\to 4^+}\frac{16-x^2}{\lvert x-4\rvert}=\lim_{x\to 4^+}\frac{(4+x)(4-x)}{(x-4)}$	**K1** (factorize)
	$=\lim_{x\to 4^+}\frac{(4+x)(4-x)}{-(4-x)}$	**K1** (simplify and substitute *x* = 4)
	$=-(4+4)$	
	$=-8$	**J1**
	TOTAL	**9 MARKS**

NO	ANSWER SCHEMES	REMARKS
2(a)	$y=\ln\left(x^2\sqrt{x-1}\right)$ $\frac{dy}{dx}=\frac{1}{x^2\sqrt{x-1}}\frac{d}{dx}\left(x^2\sqrt{x-1}\right)$ $=\frac{1}{x^2\sqrt{x-1}}\left[2x\sqrt{x-1}+\frac{x^2}{2\sqrt{x-1}}\right]$ $=\frac{1}{x^2\sqrt{x-1}}\left[\frac{4x(x-1)+x^2}{2\sqrt{x-1}}\right]$ $=\frac{x(5x-4)}{2x^2(x-1)}$ $=\frac{5x-4}{2x(x-1)}$	**K1** (correct differentiate of ln) **K1** (apply product rule) **K1** (simplify) **J1**
2(b)	$y=4\sin\left(\frac{e^x}{\ln x}\right)$ $\frac{dy}{dx}=4\cos\left(\frac{e^x}{\ln x}\right)\bullet\frac{(\ln x)\left(e^x\right)(1)-e^x\left(\frac{1}{x}\right)}{(\ln x)^2}$ $\frac{dy}{dx}=4\cos\left(\frac{e^x}{\ln x}\right)\bullet\frac{xe^x\ln x-e^x}{x(\ln x)^2}$	**K1** (correct differentiate of sin) **K1** (use formula of quotient and differentiate e^x and $\ln x$ correctly) **K1** (simplify) **J1**
	TOTAL	**8 MARKS**

NO	ANSWER SCHEMES	REMARKS
3(a)	$f(x)=3x^2-x^3$ $f'(x)=6x-3x^2$ $f'(x)=0$ $6x-3x^2=0$ $3x(2-x)=0$ $x=0, x=2$ $x=0,\ f(0)=3(0)^2-(0)^3=0$ $x=2,\ f(2)=3(2)^2-(2)^3=4$ Stationary points are (0, 0) and (2, 4)	**K1** (find $f'(x)$ and equate to 0)
	<table><tr><td></td><td>$(-\infty, 0)$</td><td>$(0, 2)$</td><td>$(2, \infty)$</td></tr><tr><td>x value</td><td>-1</td><td>1</td><td>3</td></tr><tr><td>Sign of $f'(x)$</td><td>- ve</td><td>+ ve</td><td>- ve</td></tr><tr><td>Tangent $f'(x)$</td><td>\</td><td>/</td><td>\</td></tr></table>	**K1** (any method to test)
	(0, 0) is a relative minimum point (2, 4) is a relative maximum point	**J1** (both correct)
(b)	$f(x)=3x^2-x^3$ $f''(x)=6-6x$ $f''(x)=0$ $6-6x=0$ $x=1$	**K1** (find $f''(x)$ and equate to 0)
	<table><tr><td></td><td>$(-\infty, 1)$</td><td>$(1, \infty)$</td></tr><tr><td>x value</td><td>0</td><td>2</td></tr><tr><td>Sign of $f''(x)$</td><td>+ ve</td><td>- ve</td></tr><tr><td>Tangent $f''(x)$</td><td>/</td><td>\</td></tr><tr><td></td><td>Concave upwards</td><td>Concave downwards</td></tr></table>	**K1** (any method to test)
	$x=1,\ f(1)=3(1)^2-(1)^3=2$ (1, 2) Since the function are concave upwards in the interval $(-\infty, 1)$ and concave downwards in the interval $(1, \infty)$, $\therefore$ (1, 2) is a point of inflection (shown).	**J1**

		**R1** (Shape) **R1** (All correct)
	TOTAL	**8 MARKS**

SECTION B
(*75 marks*)

NO	ANSWER SCHEMES	REMARKS
1 (a)	$z=\dfrac{(1-i)^2}{3+i}=\dfrac{1-2i+i^2}{3+i}=\dfrac{1-2i+(-1)}{3+i}$ $z=\dfrac{-2i}{3+i}\times\dfrac{3-i}{3-i}$ $z=\dfrac{-6i+2(-1)}{9-(-1)}=\dfrac{-2-6i}{10}$ $z=-\dfrac{1}{5}-\dfrac{3}{5}i$	**K1** (multiply with conjugate) **J1** (cartesian form)
1(b)	$\lvert z\rvert=\left\lvert-\dfrac{1}{5}-\dfrac{3}{5}i\right\rvert=\sqrt{\left(-\dfrac{1}{5}\right)^2+\left(-\dfrac{3}{5}\right)^2}=\dfrac{\sqrt{10}}{5}$ or 0.6325 $Arg(z)=-\left(\pi-\tan^{-1}\left\lvert\dfrac{-\frac{3}{5}}{-\frac{1}{5}}\right\rvert\right)$ $Arg(z)=-\left[\pi-\tan^{-1}(3)\right]=-1.8925\text{ rad}$	**K1** (correct formula) **K1** (correct formula) **J1** (in radian)
	TOTAL	**5 MARKS**

<table>
<tr><th>NO</th><th>ANSWER SCHEMES</th><th>REMARKS</th></tr>
<tr><td>2(a)</td><td>

$\frac{1}{3-x} \geq \frac{2}{x-7}$

$\frac{1}{3-x} - \frac{2}{x-7} \geq 0$

$\frac{x-7-2(3-x)}{(3-x)(x-7)} \geq 0$

$\frac{3x-13}{(3-x)(x-7)} \geq 0$

<table>
<tr><td></td><td>$-\infty$ to 3</td><td>3 to $\frac{13}{3}$</td><td>$\frac{13}{3}$ to 7</td><td>7 to ∞</td></tr>
<tr><td>$3x-13$</td><td>-</td><td>-</td><td>+</td><td>+</td></tr>
<tr><td>$3-x$</td><td>+</td><td>-</td><td>-</td><td>-</td></tr>
<tr><td>$x-7$</td><td>-</td><td>-</td><td>-</td><td>+</td></tr>
<tr><td>$\frac{3x-13}{(3-x)(x-7)}$</td><td>+</td><td>-</td><td>+</td><td>-</td></tr>
</table>

Solution set : $\left\{x : x < 3 \text{ or } \frac{13}{3} \leq x < 7\right\}$

</td><td>

B1
(all fraction at the left hand side and ≥ 0)

K1
(equate denominator)
K1
(correct expression)

K1
(suitable method)

K1
(choose correct interval)

J1

</td></tr>
<tr><td>2(b)</td><td>

$\left|\frac{x}{x-4}\right| < 1$

$\frac{x}{x-4} > -1$ AND $\frac{x}{x-4} < 1$

$\frac{x}{x-4} + 1 > 0$ $\quad$ $\frac{x}{x-4} - 1 < 0$

$\frac{x+x-4}{x-4} > 0$ $\quad$ $\frac{x-x+4}{x-4} < 0$

$\frac{2(x-2)}{x-4} > 0$ $\quad$ $\frac{4}{x-4} < 0$

<table>
<tr><td></td><td>$-\infty$ to 2</td><td>2 to 4</td><td>4 to ∞</td></tr>
<tr><td>$x-2$</td><td>-</td><td>+</td><td>+</td></tr>
<tr><td>$x-4$</td><td>-</td><td>-</td><td>+</td></tr>
<tr><td>$\frac{2x-4}{x-4}$</td><td>+</td><td>-</td><td>+</td></tr>
</table>

$(-\infty, 2) \cup (4, \infty)$

consider $x-4<0$

$\therefore$ $x < 4$

$(-\infty, 4)$

</td><td>

B1
(correct basic definition)

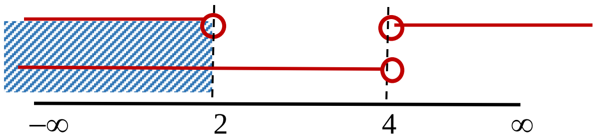
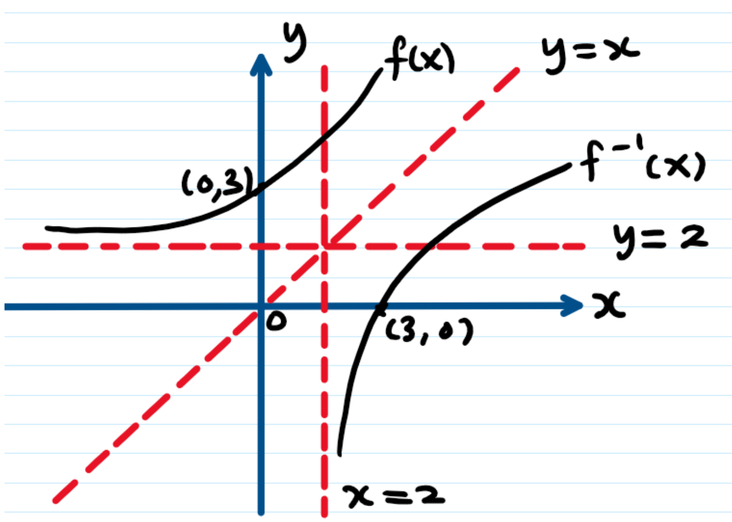
K1
(all fraction at the left hand side with either one < 0 or > 0 correct)

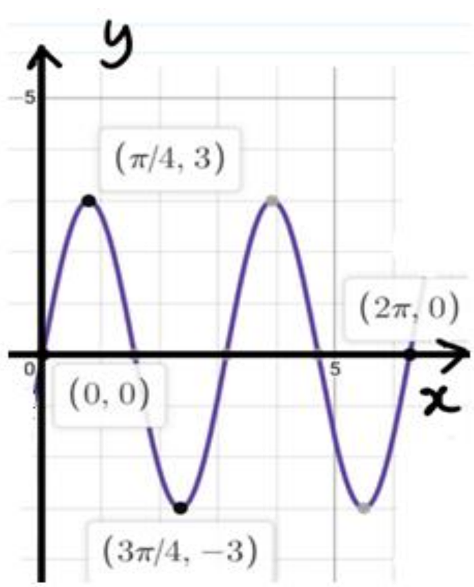
K1
(both expression correct)

K1
(correct method)

J1
(both answer correct)

</td></tr>
</table>

	 Solution set : $\{x : x < 2\}$	**K1** (Method number line) **J1**
	TOTAL	**13 MARKS**
3 (a)	$f(x) = e^{3x} + 2\ ,\ \ x \in \mathbb{R}$ $f(f^{-1}(x)) = x$ $e^{3f^{-1}(x)} + 2 = x$ $e^{3f^{-1}(x)} = x - 2$ $3f^{-1}(x) = \ln(x-2)$ $\therefore\ f^{-1}(x) = \frac{1}{3}\ln(x-2)$	**K1** (any correct method) **K1** **J1**
3(b)	 $D_f = (-\infty, \infty)$ $R_f = (2, \infty)$ $D_{f^{-1}} = (2, \infty)$ $R_{f^{-1}} = (-\infty, \infty)$	**R1** ($f\,(x)$ – shape correct) **R1** (inverse – shape correct) **R1** (both graph label, x & y axes label) **R1** (all correct with x & y intercepts) **B1** (domain & range of f) **B1** (domain & range of inverse)
	TOTAL	**9 MARKS**

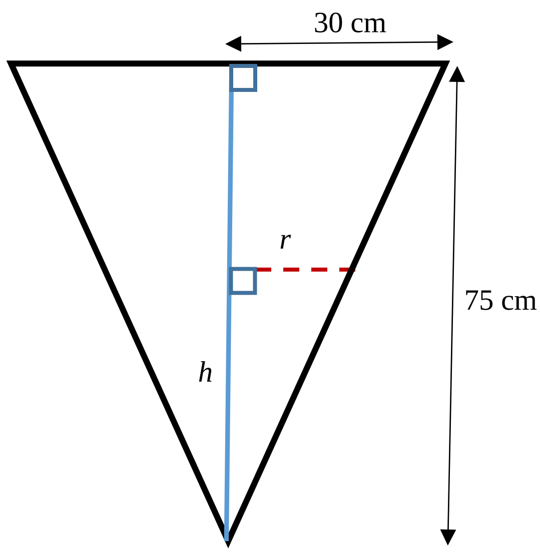
NO	ANSWER SCHEMES	REMARKS
4 (a)	$f(x)=\ln(2x-1)$ and $g(x)=2e^{-x}+1$ $g\circ f(x)=g(f(x))=g\left[\ln(2x-1)\right]$ $=2e^{-\ln(2x-1)}+1$ $=2e^{\ln(2x-1)^{-1}}+1$ $=2\left(\dfrac{1}{2x-1}\right)+1$ $g(f(x))\quad=\dfrac{2x+1}{2x-1}\quad,x\neq\dfrac{1}{2}$ $g(f(4))=\dfrac{2(4)+1}{2(4)-1}=\dfrac{9}{7}$	 **K1** **K1** **K1** **J1** **K1 J1**
4(b)	$f(x)=3\sin 2x\ ,\ 0\leq x\leq 2\pi$ $D_f=[0,2\pi]\ ,\ R_f=[-3,3]$ 	**B1** (correct domain & range) **R1** (correct shape) **R1** (All correct – *x* and *y* axes label)
	TOTAL	**9 MARKS**

NO	ANSWER SCHEMES	REMARKS
5(a)(i)	$\lim_{x\to 2}\dfrac{g(x)-g(2)}{\sqrt{2x}-2}=\lim_{x\to 2}\dfrac{g(x)-g(2)}{\sqrt{2x}-2}\times\dfrac{\sqrt{2x}+2}{\sqrt{2x}+2}$	**K1** (multiply conjugate)
	$=\lim_{x\to 2}\dfrac{\left[g(x)-g(2)\right]\left(\sqrt{2x}+2\right)}{2(x-2)}$	
	$=\lim_{x\to 2}\dfrac{g(x)-g(2)}{x-2}\bullet\lim_{x\to 2}\dfrac{1}{2}\left(\sqrt{2x}+2\right)$	
	$=g'(2)\left[\dfrac{1}{2}\left(\sqrt{2(2)}+2\right)\right]$	**K1** (simplify and substitute g'(2))
	$=3(2)$	
	$=6$	**J1**
(a)(ii)	$\lim_{x\to 2}\dfrac{1+x^2}{(x-2)(x+1)}$	
	$\lim_{x\to 2^-}\dfrac{1+x^2}{(x-2)(x+1)}=\dfrac{5}{3\lim_{x\to 2^-}(x-2)}=-\infty$	**K1**
	$\lim_{x\to 2^+}\dfrac{1+x^2}{(x-2)(x+1)}=\dfrac{5}{3\lim_{x\to 2^+}(x-2)}=+\infty$	**K1**
	$\therefore\lim_{x\to 2}\dfrac{1+x^2}{(x-2)(x+1)}$ does not exist.	**J1**
5(b) (i)	f is continuous at $x=-1$	
	$\lim_{x\to -1^-} f(x)=\lim_{x\to -1^+} f(x)$	**K1**
	$\lim_{x\to -1^-}\left(x^3-3\right)=\lim_{x\to -1^+}\left[a+(x+3)^2\right]$	**K1**
	$-4=a+4$	
	$a=-8$	**J1**
	f is continuous at $x=3$	
	$\lim_{x\to 3^-} f(x)=\lim_{x\to 3^+} f(x)$	
	$\lim_{x\to 3^-}\left[a+(x+3)^2\right]=\lim_{x\to 3^+}(bx+4)$	**K1**
	$-8+36=3b+4$	
	$28=3b+4$	
	$b=8$	**J1**

5(b)(ii)	f is not continuous at $x=6$ $\lim_{x\to 6^-} f(x) \neq \lim_{x\to 6^+} f(x)$ $\lim_{x\to 6^-}(bx+4) \neq \lim_{x\to 6^+} 8$ $6b+4\neq 8$ $6b\neq 4$ $b\neq \frac{2}{3}$	**K1** **K1** **J1**
	TOTAL	**14 MARKS**
6 (a) (i)	$x=4t+\frac{3}{t}$ $\frac{dx}{dt}=4-\frac{3}{t^2}=\frac{4t^2-3}{t^2}$ $y=4t-\frac{3}{t}$ $\frac{dy}{dt}=4+\frac{3}{t^2}=\frac{4t^2+3}{t^2}$ $\frac{dy}{dx}=\frac{\left(\frac{4t^2+3}{t^2}\right)}{\left(\frac{4t^2-3}{t^2}\right)}$ $=\frac{4t^2+3}{4t^2-3}$ By using long division $\begin{array}{r} 1 \\ 4t^2+0t-3\overline{)4t^2+0t+3} \\ \underline{4t^2+0t-3} \\ 6 \end{array}$ $\frac{dy}{dx}=1+\frac{6}{4t^2-3}\text{(shown)}$	**K1** (correct differentiate) **J1** (correct differentiate) **K1** (substitute his $\frac{dx}{dt}$ and $\frac{dy}{dt}$) **J1** **J1** (show long division)

6(a)(ii)	$\frac{d^2y}{dx^2}=\frac{d}{dt}\left(\frac{dy}{dx}\right)\times\frac{dt}{dx}$ $=-6\left(4t^2-3\right)^{-2}(8t)\left(\frac{t^2}{4t^2-3}\right)$ $=-\frac{48t^3}{\left(4t^2-3\right)^3}$ Then when $t=1, \frac{d^2y}{dx^2}=-\frac{48(1)^3}{\left(4(1)^2-3\right)^3}=-48$	**K1** (correct formula of second derivative) **K1** (correct differentiate) **K1J1**
6(b)	$f(x)=\frac{1}{\sqrt{5x+36}}=(5x+36)^{-\frac{1}{2}}$ $f'(x)=-\frac{1}{2}(5x+36)^{-\frac{3}{2}}(5)=-\frac{5}{2(5x+36)^{\frac{3}{2}}}$ $f''(x)=-\frac{5}{2}\left(-\frac{3}{2}\right)(5x+36)^{-\frac{5}{2}}(5)=\frac{75}{4(5x+36)^{\frac{5}{2}}}$ $\text{LHS}=\frac{f''(x)+f'(x)}{\sqrt{5x+36}}$ $=\frac{\frac{75}{4(5x+36)^2\sqrt{5x+36}}-\frac{5}{2(5x+36)\sqrt{5x+36}}}{\sqrt{5x+36}}$ $=\frac{75}{4(5x+36)^3}-\frac{5}{2(5x+36)^2}$ $=\frac{75}{4}\left[\frac{1}{(5x+36)^{\frac{1}{2}}}\right]^6-\frac{5}{2}\left[\frac{1}{(5x+36)^{\frac{1}{2}}}\right]^4$ $=\frac{75}{4}[f(x)]^6-\frac{5}{2}[f(x)]^4$ $=\text{RHS}$ (shown)	**K1** (correct differentiate) **K1** (correct differentiate) **K1** (substitute his f' and f'') **K1** (simplify) **K1** **J1**
	TOTAL	**15 MARKS**

7 (a)



$$\frac{dV}{dt} = 180\ cm^3 s^{-1}$$

$$\frac{r}{h} = \frac{30}{75} \Rightarrow r = \frac{2}{5}h$$

K1 (*r* as a subject)

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi\left(\frac{2}{5}h\right)^2 h$$

$$V = \frac{4}{75}\pi h^3$$

K1 (find *V* in terms of *h*)

$$\frac{dV}{dh} = 3\left(\frac{4}{75}\right)\pi h^2 = \frac{4}{25}\pi h^2$$

K1 (differentiate *V* wrt *h*)

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} = \frac{25}{4\pi h^2} \times 180$$

K1 (apply chain rule)

$$\frac{dh}{dt} = \frac{1125}{\pi h^2} \quad \text{(shown)}$$

J1 (as it is)

7(b)	$r=\frac{2}{5}h \Rightarrow \frac{dr}{dh}=\frac{2}{5}$ $\frac{dr}{dt}=\frac{dr}{dh}\times\frac{dh}{dt}=\frac{2}{5}\times\frac{1125}{\pi h^2}$ $\frac{dr}{dt}=\frac{450}{\pi h^2}$	**K1** (find $\frac{dr}{dt}$)
	when $r=\frac{1}{3}(30)=10$ cm, $h=\frac{5}{2}r=\frac{5}{2}(10)=25$ cm $\therefore \frac{dr}{dt}=\frac{450}{\pi h^2}=\frac{450}{\pi(25)^2}=\frac{18}{25\pi}$ cm/s	**K1J1**
	ALTERNATIVE METHOD $\frac{r}{h}=\frac{30}{75} \Rightarrow h=\frac{75}{30}r$ $V=\frac{1}{3}\pi r^2 h=\frac{1}{3}\pi r^2\left(\frac{75}{30}r\right)=\frac{75}{90}\pi r^3$ $\frac{dV}{dr}=\frac{75}{30}\pi r^2$ $\frac{dr}{dt}=\frac{dr}{dV}\times\frac{dV}{dt}=\frac{30}{75\pi r^2}\times 180$ $\frac{dr}{dt}=\frac{72}{\pi r^2}$ when $r=\frac{1}{3}(30)=10$ cm, $\therefore \frac{dr}{dt}=\frac{72}{\pi r^2}=\frac{72}{\pi(10)^2}=\frac{18}{25\pi}$ cm	
7(c)	$\frac{dV}{dt}=180 \text{ cm}^3 s^{-1} \Rightarrow V=180\,t$ Volume, $V=\frac{1}{3}\pi(30)^2(75)=22500\pi \text{ cm}^3$	
	$180\,t=22500\pi$	**K1**
	$t=\frac{22500\pi}{180\times 60}=6.54$ minutes	**J1**
	TOTAL	**10 MARKS**